Hybrid Inheritance

In [ ]:
```
Hybrid inheritance is a
combination of two or more types of inheritance in a single program

Example:

Multiple+Multilevel
Hierarchical+ Multiple
```

In [ ]:
```
Syntax:

class A:
    #Logic

class B(A):
    #Logic

class C(A):
    #Logic

class D(B,C):
    #LOGIC

MRO  -->Method resolution order


D  --->B--->C--->A--->object

print(D.mro)
```

University Management system

In [ ]:
```
           Person
Teacher                Researcher
           Professor
```

In [27]:
```
class Person:
    def __init__(self,name):
        self.name=name
        print("Person class constructor!")

class Teacher(Person):
    def __init__(self,name,subject):
        Person. __init__(self,name)
        self.subject=subject
        print("I am from Teacher class...")

class Researcher(Person):
    def __init__(self,name,research_area):
        Person. __init__(self,name)
        self.research_area=research_area
        print("I am from Research class!")

class Professor(Teacher,Researcher):
    def __init__(self,name,subject,research_area):
        Teacher.__init__(self,name,subject)
        Researcher.__init__(self,name,research_area)
        print("I am from Professor class...")

    def display(self):
        print("Name: ",self.name)
        print("Subject: ",self.subject)
        print("Research area: ",self.research_area)

p=Professor('Dr.Amit','Artificial Intelligence','GenAI')
```

```
Person class constructor!
I am from Teacher class...
Person class constructor!
I am from Research class!
I am from Professor class...
```

In [29]:
```
p.display()
```

```
Name:  Dr.Amit
Subject:  Artificial Intelligence
Research area:  GenAI
```

In [25]:
```
for i in (Professor.mro()):
    print(i)
```

```
<class '__main__.Professor'>
<class '__main__.Teacher'>
<class '__main__.Researcher'>
<class '__main__.Person'>
<class 'object'>
```

In [ ]:
```
                   order(o_id)

Payment(o_id,amount)          Shipping(o_id,address)

       uOrder(o_id,amount,address)
```

In [ ]:
```
Smart Home System

        SmartDevice(BRAND)

Camera(brand,resolution)              Speaker(brand,watts)

        SmartTV(screen_size)
```